

| Question |     |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks available |     |     |       |       |      |
|----------|-----|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |     |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 8/2      | (a) |  |  | 0.42 (2)<br><br>if answer incorrect award (1) for any of following<br>42<br>420000<br>$4.2 \times 10^5$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                 | 2   |     | 2     | 2     |      |
|          | (b) |  |  | award (1) for any of following <ul style="list-style-type: none"><li>• distillation</li><li>• reverse osmosis</li><li>• graphene membrane</li></ul> neutral answers<br>desalination<br>boiling / evaporation / condensation                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 1               |     |     | 1     |       |      |
|          | (c) |  |  | add soap solution to both water samples <b>and</b> shake (1)<br><br>fair testing - <b>equal volumes</b> of both water samples <b>and</b> <b>shake for an equal amount of time</b> (1)<br><br>the sample that produces more lather is the softer water / the sample that produces less lather is the harder (1)<br><br>alternative method<br><br>add 1cm <sup>3</sup> / small volume of soap solution and shake; if no lather add further 1cm <sup>3</sup> / small volume (1)<br><br>fair testing - <b>equal volumes</b> of both water samples <b>and</b> <b>shake for an equal amount of time</b> (1)<br><br>the sample that lathers with less soap is the softer / the sample that requires more soap is the harder (1) | 3               |     |     | 3     |       | 3    |

| Question |  |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                         | Marks available |     |     |       |       |      |
|----------|--|-------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |       |  |                                                                                                                                                                                                                                                                                                                                                                         | AO1             | AO2 | AO3 | Total | Maths | Prac |
|          |  | (iii) |  | <div><div><math>2\text{Ca} + \text{O}_2 \longrightarrow 2\text{CaO}</math></div><div><math>\text{Ca} + \text{O}_2 \longrightarrow 2\text{CaO}</math></div><div><math>\text{Ca} + \text{O} \longrightarrow \text{CaO}</math></div></div> <div><div><input checked="" type="checkbox"/></div><div><input type="checkbox"/></div><div><input type="checkbox"/></div></div> |                 | 1   |     | 1     |       |      |
|          |  |       |  | Question 1 total                                                                                                                                                                                                                                                                                                                                                        | 0               | 8   | 0   | 8     | 0     | 0    |